

Size as an Intrinsic Landmark: How Icon Size Variation Supports Spatial Learning in Graphical Interfaces

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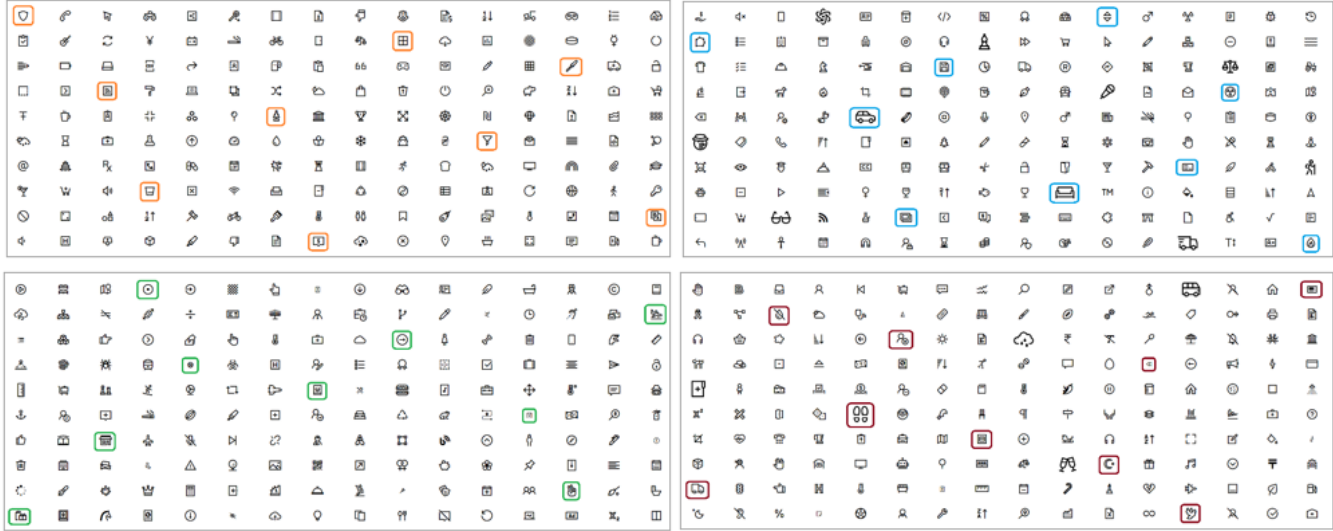


Figure 1: Interface variants used in the study. Target icons are highlighted in color. Top-left: Baseline; top-right: Baseline+Large; bottom-left: Baseline+Small; bottom-right: Baseline+Mixed.

Abstract

Large graphical user interfaces expose hundreds of commands, yet users often rely on repeated visual search instead of recall. While spatial memory can support expert performance, dense and visually uniform command grids provide few anchors for forming stable spatial representations. We investigate whether icon size variation can serve as a subtle, intrinsic visual landmark that supports spatial learning without adding external cues or cluttering the interface. In a repeated-selection study ($N = 16$) using a spatially stable 160-icon grid, we compare a uniform baseline with interfaces containing a small number of enlarged, reduced, or mixed-size icons. Results show that size variation improves performance during later recall blocks while maintaining low error rates. We discuss implications for designing dense command interfaces and outline future directions for combining intrinsic visual cues to aid spatial learning in more realistic command systems.



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1 Introduction

Modern graphical user interfaces (GUIs) increasingly expose large command sets through toolbars, ribbons, and grid-based menus. While selecting a command is a fundamental interaction, efficiency depends on users' ability to learn and recall command locations rather than repeatedly relying on visual search [9, 15]. As the number of commands increases, however, users often struggle to form stable spatial representations, resulting in slower performance and greater cognitive effort [1, 2, 4].

Prior work shows that spatially stable layouts can support the transition from novice, search-driven behavior to expert, memory-driven interaction [9, 15, 32]. Yet many large command interfaces remain visually uniform, offering few distinctive cues to help users organize and recall locations, particularly away from natural landmarks such as edges and corners [26]. Research in spatial cognition highlights the role of landmarks—stable, distinctive reference points—in structuring spatial memory [22, 25, 28]. In dense graphical interfaces, however, such landmarks are often absent, limiting the effectiveness of spatial learning as interfaces scale.

To address this, prior systems have introduced artificial landmarks, such as background images, colored regions, or overlays, to support spatial memory [6, 7, 28, 32]. While effective, these approaches add external visual elements that may clutter already dense interfaces and interfere with usability [24]. An alternative is to leverage intrinsic visual properties of interface elements themselves as landmarks, avoiding additional visual layers.

In this work, we investigate whether icon size variation can serve as an intrinsic visual landmark that supports spatial learning without altering layout stability or interaction mechanics. Size is a perceptually salient visual variable [10, 35], making it a promising candidate for introducing distinctiveness within otherwise uniform grids. Importantly, size variation introduces no new interface elements, preserving visual simplicity.

We evaluate this approach in a repeated-selection study using a spatially stable 160-icon grid with semantically meaningful icons. By leveraging icon size as an intrinsic landmark, this work examines a lightweight design strategy for supporting spatial learning in large command interfaces without introducing additional visual elements. The findings inform the design of scalable GUIs by identifying intrinsic visual cues that better support the transition from search-driven to memory-driven interaction.

2 Related Work

Prior work shows that spatially stable interfaces can support the transition from novice, search-driven interaction to expert, memory-driven use by enabling durable associations between commands and their locations [8, 12, 18]. A range of memory-based interaction techniques leverage this principle, including radial menus [11, 20] and grid-based command interfaces such as CommandMaps, ListMaps, and FastTap [15, 16, 27]. However, as command density increases, spatial memory benefits often diminish, particularly in visually uniform layouts [32].

Landmarks play a critical role in structuring spatial memory in both physical and digital environments [22, 26]. In GUIs, natural landmarks such as screen edges and corners can support recall of nearby items, but their influence weakens in large displays where many commands are distant from these regions [30, 31]. To address this limitation, prior work has introduced artificial landmarks, including background images, colored regions, and overlays, to reinforce spatial memory [3, 5, 6, 32, 33]. However, such approaches rely on additional visual elements, which motivates interest in intrinsic landmarking strategies.

Size is a fundamental visual variable that guides attention and is processed pre-attentively [17, 29, 35, 36]. Prior HCI research has leveraged size variation to improve recognition-based performance

in GUIs [12, 21], yet its role in supporting recall remains underexplored. This distinction is particularly relevant in light of models of skill acquisition, which describe a progression from search-driven recognition to memory-based recall as users gain experience [14]. Building on this gap, our work examines whether icon size variation can function as an intrinsic landmark in spatially stable interfaces, supporting spatial learning without introducing external cues.

3 Study: Icon Size Effects on Spatial Memory

3.1 Study Interfaces

We designed four grid-based command interfaces inspired by CommandMaps [27], which display all commands simultaneously while preserving spatial stability (Figure 1). Each interface contained 160 semantically meaningful grayscale icons selected from the *Line Awesome* icon set [19], arranged in a 10×16 grid. Icon identities and spatial positions were fixed across all conditions. The grid occupied most of the display, with a reserved region at the bottom for target labels.

The baseline icon size was 45 px. In the Baseline interface, all 160 icons were rendered at this size. In the Baseline+Large interface, 10 icons were enlarged to 80 px, while the remaining icons retained their baseline size; in the Baseline+Small interface, 10 icons were reduced to 25 px. The Mixed interface combined both manipulations, with 5 icons enlarged to 80 px and 5 reduced to 25 px. In all size-variation interfaces, the size-modified icons were distributed across the grid to avoid clustering. Only the rendered icon size varied across conditions; *clickable areas were identical for all icons* to avoid confounding pointing difficulty. Size values were chosen based on prior design guidelines and iterative inspection to ensure recognizability and perceptual salience in a dense layout.

We manipulated 10 of 160 icons (6.25%) to keep landmarking sparse and avoid clutter while still providing salient anchors. Landmark density (i.e., the proportion and placement of size-modified icons) is a potential design parameter that we plan to explore in follow-up studies.

3.2 Methodology

3.2.1 Task and Procedure. Participants performed a repeated target-selection task. At the start of each trial, a target label was shown at the bottom of the screen, and participants selected the corresponding icon from the grid using a mouse. Visual feedback was provided after each selection, and participants could proceed only after a correct response.

For each interface, participants completed 12 blocks of 9 trials, using the same set of target icons presented in randomized order. Blocks 1–3 were treated as *learning* blocks, and Blocks 4–12 as *recall* blocks. The study followed a within-participants design, and the order of interface conditions was counterbalanced using a balanced Latin square. Short breaks were allowed between conditions to reduce fatigue.

3.2.2 Participants and Apparatus. Sixteen participants (8 women, 8 men), aged 24–47 years ($M = 31.5$), were recruited from a university population. All participants reported regular experience with desktop graphical user interfaces. Each session lasted approximately 60 minutes, and participants received CA\$10 compensation.

The experiment was conducted on a desktop computer running Windows 11 Pro, equipped with a 27-inch monitor (1920×1080 resolution). Participants were seated at an approximate viewing distance of 60 cm. Under these conditions, the reduced (25 px), baseline (45 px), and enlarged (80 px) icons subtended approximately 0.7°, 1.2°, and 2.1° of visual angle, respectively. Input was provided using a standard optical mouse and keyboard. All instructions and target labels were presented in English.

3.3 Results

We measured completion time (ms) and error rate (incorrect selections). Data were analyzed using repeated-measures ANOVAs across all blocks combined and separately for the learning and recall phases. Effect sizes are reported as general eta-squared (η^2) and interpreted using standard thresholds (small: .01, medium: .06, large: .14; [13]).

3.3.1 Completion Time Analysis. Table 1 summarizes completion times. Completion time decreased substantially over blocks across all interfaces, indicating a strong practice effect. While performance was similar during learning, size-variant interfaces achieved lower completion times than the Baseline during recall. ANOVA results (Table 2) show a strong main effect of Block and a significant Interface×Block interaction, indicating that interface differences emerged with practice. Because our study consisted of a single session, we describe recall-phase separation primarily as a performance pattern that emerges with practice rather than evidence of long-term skill consolidation. As shown in Figure 2, all interfaces improved rapidly during learning and stabilized during recall, with Baseline+Large and Baseline+Small reaching lower asymptotic completion times from Block 4 onward.

Table 1: Completion time (ms) by interface. Values are means with standard deviations shown beneath.

Interface	All Blocks	Learning Blocks	Recall Blocks
Baseline	5216.06 (11512.09)	11511.56 (19081.71)	3117.57 (6158.49)
Baseline+Large	4628.28 (10904.43)	11814.25 (19451.10)	2232.96 (3111.99)
Baseline+Small	5062.10 (15222.87)	13865.65 (28308.93)	2127.58 (2801.63)
Baseline+Mixed	4801.50 (12048.66)	11449.07 (21627.51)	2585.65 (4273.96)

3.3.2 Error Analysis. Error rates were low across all interfaces and declined with practice (Table 3). Errors were slightly higher during early learning but near zero during recall (Figure 3). ANOVA results (Table 4) show a significant Block effect and a transient Interface×Block interaction, with no recall-phase interface differences, suggesting that completion time differences are unlikely to reflect speed–accuracy trade-offs.

3.3.3 Interface Position Analysis. Table 5 indicates that targets near edges and corners were selected faster than central targets, with

Table 2: Completion time ANOVA results.

Effect	All Blocks			Learning Blocks			Recall Blocks		
	F	p	η^2	F	p	η^2	F	p	η^2
Interface	0.55	.65	.005	1.40	.25	.030	3.44	<.05	.06
Block	199.43	<.001	.766	215.61	<.001	.687	6.32	<.001	.08
Interface × Block	2.74	<.001	.102	3.04	.009	.094	1.41	.10	.05

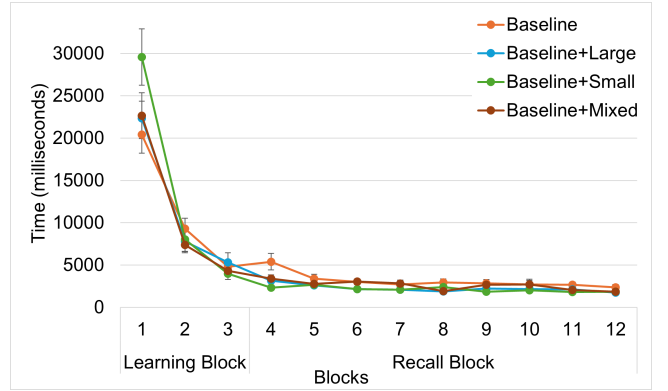


Figure 2: Mean Completion time across blocks for all interfaces.

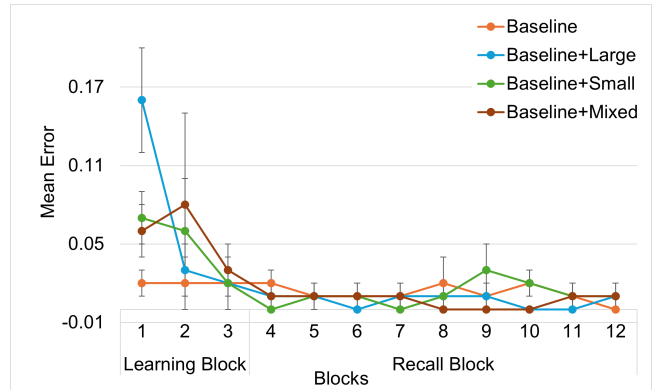


Figure 3: Mean error rate across blocks for all interfaces.

Table 3: Error rate: mean (SD).

Interface	All Blocks	Learning Blocks	Recall Blocks
Baseline	0.02 (0.14)	0.02 (0.17)	0.01 (0.13)
Baseline+Large	0.02 (0.19)	0.07 (0.35)	0.01 (0.08)
Baseline+Small	0.02 (0.20)	0.05 (0.35)	0.01 (0.12)
Mixed	0.02 (0.27)	0.06 (0.52)	0.01 (0.09)

this advantage apparently consistent in recall (though marginal in recall-only analysis). ANOVA revealed a significant effect of position and a Position×Block interaction (Table 6), indicating that positional benefits strengthened with practice (Figure 4). These

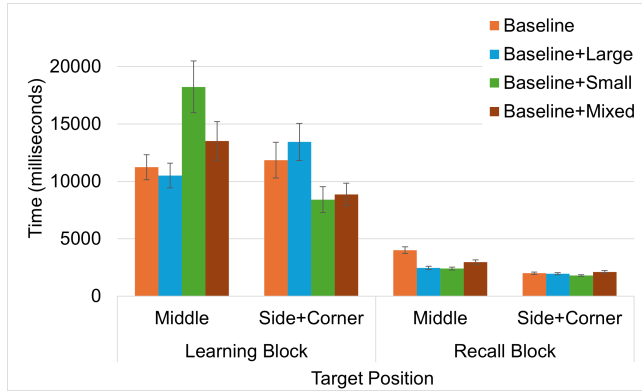
Table 4: Error rate ANOVA results.

Effect	All Blocks			Learning Blocks			Recall Blocks		
	F	p	η^2	F	p	η^2	F	p	η^2
Interface	1.06	.37	.002	2.87	.047	.033	1.20	.32	.01
Block	4.71	<.001	.100	3.16	.057	.050	0.42	.91	.01
Interface $\times$ Block	2.00	.001	.075	2.21	.050	.078	0.88	.63	.04

results align with natural anchors that aid in recall, suggesting that intrinsic landmarks, such as variations in icon size, may help reduce spatial learning disadvantages in the central region.

Table 5: Completion time (ms) by interface position. Values are means with standard deviations shown beneath.

Position	All Blocks	Learning Blocks	Recall Blocks
Side+Corner	4131.87 (10273.08)	10639.96 (18749.58)	1962.51 (2194.79)
Middle	5563.08 (14050.70)	13376.27 (24926.96)	2958.69 (5397.94)

**Figure 4: Completion time by target position (middle vs. side+corner) during learning and recall blocks (mean $\pm$ SE).****Table 6: Interface position ANOVA results (completion time).**

Effect	All Blocks			Learning Blocks			Recall Blocks		
	F	p	η^2	F	p	η^2	F	p	η^2
Position	4.92	.008	.021	3.67	.031	.019	2.88	.060	.015
Block	199.43	<.001	.766	215.61	<.001	.687	6.32	<.001	.081
Interface $\times$ Block	1.84	.002	.041	1.53	.048	.037	1.21	.25	.021

NASA-TLX ratings showed no significant differences in workload scores across interfaces, indicating no additional subjective effort. Participants' preferences strongly favored the Baseline+Large

interface, with eight or more of the sixteen participants identifying it as their preferred interface for Speed, Memorization, Expertise development, and Overall measures.

3.3.4 Scope and Limitations. It is worth noting that our controlled, single-session study ($N = 16$) prioritizes internal validity and effect-size reporting but limits generalizability to the population. Ecological validity is constrained by the static grid; real command interfaces often include grouping, labels, and color that may interact with intrinsic size cues. We position size variation as a lightweight, incremental intrinsic landmark that can complement existing organization and landmarking strategies.

4 Discussion

This study examined whether variation in icon size can serve as an intrinsic visual landmark to support spatial learning in dense command interfaces. Results show that size-variant interfaces yielded faster recall-phase performance without increasing error rates, suggesting improved recall-phase performance after initial learning.

Completion times exhibited a strong practice effect across all interfaces, with rapid early improvement followed by stabilization. Crucially, size-variant interfaces converged to lower recall-phase completion times than the uniform baseline, reflected by a significant Interface $\times$ Block interaction. This pattern is consistent with prior work showing that stable spatial cues facilitate the transition from search-driven to recall-driven interaction with practice [14, 23].

Error rates were low across all conditions and showed no recall-phase differences between interfaces, suggesting that performance gains in size-variant interfaces do not reflect speed-accuracy trade-offs but more efficient retrieval of spatially learned command locations.

Interface position analysis further highlights the role of spatial structure. Targets near corners and sides were selected faster than central targets during recall, consistent with geometric boundaries acting as natural spatial anchors [12, 34]. The persistence of size-based benefits across interfaces suggests that intrinsic landmarks can complement such anchors, supporting spatial organization even in anchor-poor regions.

Overall, these findings indicate that icon-size variation can provide a lightweight, intrinsic landmarking strategy that can support the shift toward recall-based performance in later blocks without adding visual clutter or disrupting layout stability.

5 Conclusion and Future Work

This work demonstrates that icon size variation can serve as an intrinsic visual landmark that supports spatial learning in dense command interfaces. In a repeated-selection task using a spatially stable grid of semantic icons, size variation reduced recall-phase completion times without increasing error rates, offering a simple, low-cost mechanism for supporting memory-driven interaction.

The study used a controlled task and a fixed grid layout. Real-world interfaces include additional cues such as semantic grouping, labels, and task context, which may interact with intrinsic landmarking. Future work could examine longer-term retention of commands in spatial memory (preferably across multiple sessions), variable landmark density, and the deployment of these strategies in

real command systems to assess ecological validity and scalability. We also plan to evaluate size cues alongside other intrinsic GUI features such as grouping, labels or colors in more realistic command interfaces. As CHI is a broad community, we aim to discuss these findings with the audience to gather recommendations for extending this line of work.

Acknowledgments

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